

Fano 2-22 is the Feynman graphon: emergence of FLRW spacetime from a no-free-parameter algebraic construction

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Abstract

We construct a sequence of weighted regular graphs $\{G_k\}$ whose spectral data are derived from the Minkowski period coefficients of the Fano 3-fold 2–22 (Mori-Mukai ID-69). The coefficients factorize as

$$c_5 = 24 \times 45, \quad c_6 = \left(\frac{4}{3} \times 45\right) \times (45 + 64), \quad c_7 = 24 \times \left(\frac{4}{3} \times 45\right) \times (45 - 10),$$

which we interpret as trace invariants (spectral moments) of the adjacency matrix of a base graph G_{45} with $d_{\text{adj}} = 45$ vertices and degree $\deg(G) = 24$. The topological volume of the Fano 2–22 cell is $V_{\text{cell}} = \delta_{\text{th}}$, where $\delta_{\text{th}} = \frac{1}{2\pi} \ln(\sqrt{8}/(\sqrt{8} - \sqrt{7}))$. A regular lattice of such cells filling the 3-sphere of radius $R = \pi$ yields a discrete spacetime whose continuum limit is the FLRW 3-sphere.

We prove convergence in cut norm of the graph sequence $\{G_k\}$ to the limit graphon $W(x, y)$ with rate $\mathcal{O}(2^{-k})$, where the kernel and its parameters are defined by the compact collection

$$\begin{aligned} W(x, y) &= \kappa \Phi_{\text{VEV}}^2 e^{-\gamma d(x, y)}, \\ \Delta S &= 2\sqrt{2} - \sqrt{7}, \\ \kappa &= \frac{24}{25} \cdot \frac{\delta_{\text{th}}}{\Delta S}, \\ \gamma &= \sqrt{\pi^4 + 1} \cdot \frac{24}{25} \cdot \frac{\delta_{\text{th}}}{\Delta S}. \end{aligned}$$

We derive the exact spectral gap of the graphon via spherical harmonic expansion on S^3 , expressing the eigenvalues in terms of Gegenbauer polynomials. Numerical evaluation yields $\lambda_1/\lambda_0 \approx 1.99765$, which asymptotically tends to 2 as $\gamma R \rightarrow \infty$, reflecting the isotropy and homogeneity of the emergent spacetime. The construction contains no free parameters and provides a purely geometric realization of spacetime emergence from a discrete combinatorial structure.

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Notation and conventions

Symbol	Definition
π	The circle constant, 3.141592653589793...
R	Radius of the 3-sphere, fixed to $R = \pi$
$SO(10)$	Gauge group of the 5D theory
$\mathfrak{so}(10)$	Lie algebra of $SO(10)$
$d_{\text{adj}} = 45$	Dimension of the adjoint representation of $SO(10)$
Φ	Scalar field in the adjoint representation
Φ_{VEV}^2	Vacuum expectation value, $\pi^4 + 1$
Φ_{VEV}	$\sqrt{\pi^4 + 1} \approx 9.9201356359$
ΔS	Entanglement deficit, $2\sqrt{2} - \sqrt{7}$
δ_{th}	Picard-Fuchs threshold, $\frac{1}{2\pi} \ln \left(\frac{\sqrt{8}}{\sqrt{8}-\sqrt{7}} \right)$
γ	Geometric threshold, $\sqrt{\pi^4 + 1} \cdot \frac{24}{25} \cdot \frac{\delta_{\text{th}}}{\Delta S} \approx 22.732171$
κ	Contact coupling, $\frac{24}{25} \cdot \frac{\delta_{\text{th}}}{\Delta S}$
I_h	Holonomy invariant, $4/(3\pi)$
πI_h	Hinge factor, $4/3$
T	String tension, $T = 8$
24	Number of transverse modes
10	Dimension of $SO(5)$ and of $\mathfrak{so}(4, 1)$
$\deg(G) = 24$	Degree of the regular graph G_{45}
G_{45}	Base graph with $d_{\text{adj}} = 45$ vertices
G_k	Graph at refinement level k (for $k = 0, 1, 2, \dots$)
N_k	Number of vertices of G_k (scales as $2^k \cdot N_0$)
A	Adjacency matrix of a graph
$W(x, y)$	Graphon kernel
$d(x, y)$	Geodesic distance on S^3
μ	Haar measure on S^3
$Y_{\ell, a}$	Spherical harmonics on S^3 with degree ℓ and index a

Symbol	Definition
$g_\ell = (\ell + 1)^2$	Degeneracy of spherical harmonics of degree ℓ
$C_\ell^{(1)}$	Gegenbauer polynomial of degree ℓ and order 1
$\delta_\square$	Cut distance
$\text{Inj}(S^3) = \pi$	Injectivity radius of S^3
$\mathcal{O}$	Big-O asymptotic notation
$\widehat{\mathcal{M}}$	Tensor density operator (holographic measure)
Π	Projection onto $SO(5)$ -orthogonal subspace

1 Introduction and motivation

The question of how continuous spacetime emerges from discrete combinatorial structures is a central open problem in mathematics and theoretical physics. A recent development in graph theory [4] describes the continuum limit of large dense graphs via **graphons** (limits of dense graph sequences), with the spectral gap of the adjacency matrix providing the threshold between the discrete and continuous phases.

The Fano 3-fold 2–22 (Mori–Mukai ID-69) has appeared in previous work [2] as the algebraic moduli space of the PEPS-5D Lagrangian. Its Minkowski period coefficients exhibit factorizations involving the numbers $d_{\text{adj}} = 45$, 24, $4/3$, 64, and 10, which are precisely the structural parameters of the gauge theory: the bond dimension, the transverse modes of the string, the hinge factor $\pi I_h = 4/3$, the square of the string tension $T^2 = 64$, and the dimension of $\mathfrak{so}(4, 1)$.

In this work we prove that the Fano 2–22 generates a sequence of graphs converging to a graphon whose continuum limit yields the 4D Lagrangian derived from the PEPS-5D construction. We construct the graph G_{45} explicitly, prove convergence in cut norm, and derive the exact spectral gap of the limiting graphon via spherical harmonic analysis on S^3 . The appendix provides three rigorous theorems showing that the discrete metric spaces converge (Gromov–Hausdorff) to S^3 , the discrete Regge action converges to the Einstein–Hilbert functional, and the discrete temporal extension yields the Friedmann equation — all with no free parameters.

2 The 5D gauge theory and holographic reduction

We recall the 5D gauge theory constructed in [2]. The gauge group is $G = SO(10)$, with the adjoint representation dimension

$$\dim(\text{adj } SO(10)) = \frac{10 \cdot 9}{2} = d_{\text{adj}} = 45. \quad (1)$$

The 5D Lagrangian is

$$\mathcal{L}_{5\text{D}} = -\frac{1}{4g^2} \text{Tr}(F_{MN}F^{MN}) + \frac{1}{2} \text{Tr}(D_M \Phi D^M \Phi) - \frac{\lambda}{4} (\text{Tr}(\Phi^2) - (\pi^4 + 1))^2 + \mathcal{L}_{\text{hinge}}, \quad (2)$$

where

$$\mathcal{L}_{\text{hinge}} = \Phi^2 |e^{-i\pi/6} \Psi_d - I_h \Psi_\Phi|^2, \quad I_h = \frac{4}{3\pi}. \quad (3)$$

The fifth dimension is projective, not spatial. Integration over the projective phase θ with range $\Delta\theta = \pi/3$ (spinorial doubling of the hinge angle $\pi/6$) yields

$$\int_0^{\pi/3} \mathcal{L}_{\text{hinge}} d\theta = \kappa \Phi^2 \delta^{(3)}(\vec{x}), \quad (2.1)$$

where $\delta^{(3)}(\vec{x})$ is the contact density localized at each lattice node before the continuum limit. In the homogeneous limit, this contact term is smeared into the constant measure. The geometric invariants are

$$\Delta S = 2\sqrt{2} - \sqrt{7}, \quad (5)$$

$$\delta_{\text{th}} = \frac{1}{2\pi} \ln \left(\frac{\sqrt{8}}{\sqrt{8} - \sqrt{7}} \right), \quad (6)$$

$$\kappa = \frac{24}{25} \cdot \frac{\delta_{\text{th}}}{\Delta S}. \quad (7)$$

2.1 Three forms of the 4D Lagrangian

The effective 4D Lagrangian after holographic reduction can be written in three equivalent forms.

Expanded form (explicit invariants):

$$\begin{aligned} \mathcal{L}_{4\text{D}}^{(\text{exp})} = & -\frac{1}{4g^2} \text{Tr}(F_{\mu\nu}F^{\mu\nu}) \\ & + \frac{1}{2} \text{Tr}(D_\mu \Phi D^\mu \Phi) \\ & - \frac{\lambda}{4} (\text{Tr}(\Phi^2) - (\pi^4 + 1))^2 \\ & + \frac{24}{25} \cdot \frac{1}{2\pi} \ln \left(\frac{\sqrt{8}}{\sqrt{8} - \sqrt{7}} \right) \cdot \frac{\Phi^2}{2\sqrt{2} - \sqrt{7}} \cdot \delta^{(3)}(\vec{x}). \end{aligned}$$

Compact form (contact term):

$$\mathcal{L}_{4\text{D}}^{(\text{comp})} = -\frac{1}{4g^2} \text{Tr}(F_{\mu\nu}F^{\mu\nu}) + \frac{1}{2} \text{Tr}(D_\mu \Phi D^\mu \Phi) - \frac{\lambda}{4} (\text{Tr}(\Phi^2) - (\pi^4 + 1))^2 + \kappa \Phi^2 \delta^{(3)}(\vec{x}),$$

Renormalized form (absorbed couplings):

$$\mathcal{L}_{4\text{D}}^{(\text{ren})} = -\frac{1}{4g_{\text{eff}}^2} \text{Tr}(F_{\mu\nu}F^{\mu\nu}) + \frac{1}{2} \text{Tr}(D_\mu \Phi D^\mu \Phi) - \frac{\lambda_{\text{eff}}}{4} (\text{Tr}(\Phi^2) - (\pi^4 + 1))^2,$$

with

$$g_{\text{eff}}^2 = g^2 \cdot \left(\frac{24}{25} \cdot \frac{\delta_{\text{th}}}{\Delta S} \right)^{-1}, \quad \lambda_{\text{eff}} = \lambda \cdot \frac{24}{25} \cdot \frac{\delta_{\text{th}}}{\Delta S}. \quad (11)$$

2.2 The geometric threshold

The geometric threshold (a pure number, dimensionless) is given by

$$\gamma = \sqrt{\pi^4 + 1} \cdot \frac{24}{25} \cdot \frac{\delta_{\text{th}}}{\Delta S} \approx 22.732171. \quad (12)$$

This number determines the exponential decay scale of the graphon kernel and appears in the spectral gap formula derived in section 9.

3 The Fano 2–22 and its combinatorial data

The Fano 3-fold 2–22 (Mori-Mukai ID-69) has Minkowski period sequence [3]

$$\widehat{G}_X(t) = 1 + 6t^2 + 24t^3 + 138t^4 + 1080t^5 + 6540t^6 + 50400t^7 + \dots \quad (13)$$

Let c_n denote the coefficient of t^n . The coefficients factorize exactly as [2]:

$$c_5 = 24 \times d_{\text{adj}} = 1080, \quad (14)$$

$$c_6 = \left(\frac{4}{3} \times d_{\text{adj}}\right) \times (d_{\text{adj}} + 64) = 6540, \quad (15)$$

$$c_7 = 24 \times \left(\frac{4}{3} \times d_{\text{adj}}\right) \times (d_{\text{adj}} - 10) = 50400. \quad (16)$$

We interpret these coefficients as **spectral moments** of the adjacency matrix A of a graph G_{45} with $d_{\text{adj}} = 45$ vertices:

$$\text{Tr}(A^2) = c_5, \quad \text{Tr}(A^3) = c_6, \quad \text{Tr}(A^4) = c_7. \quad (17)$$

Remark 3.1 (Spectral moment problem and semidefinite programming). *The identification of the Minkowski coefficients c_5, c_6, c_7 with the spectral moments $\text{Tr}(A^2), \text{Tr}(A^3), \text{Tr}(A^4)$ is nontrivial and requires solving a truncated spectral moment problem. Concretely, we solve the following semidefinite program (SDP): find a symmetric matrix $A \in \mathbb{R}^{45 \times 45}$ minimizing 0 subject to*

$$A_{ii} = 0, \quad A_{ij} \geq 0, \quad \sum_j A_{ij} = 24 \quad \forall i,$$

$$\text{Tr}(A^2) = c_5, \quad \text{Tr}(A^3) = c_6, \quad \text{Tr}(A^4) = c_7.$$

Feasibility of this SDP yields a weighted regular adjacency matrix realizing the moments. Numerical feasibility (Appendix B) produces an explicit rational matrix A satisfying the constraints to machine precision; the existence statement may be promoted to a theorem conditional on SDP feasibility. See Curto–Fialkow for the truncated moment problem background.

The factorization (14) forces a regular graph of degree $\deg(G) = 24$. The third moment gives the number of triangles $\Delta = 1090$. The fourth moment involves the subtraction $d_{\text{adj}} - 10$, indicating a projection onto the subspace orthogonal to a 10-dimensional subalgebra. This projection is encoded in the **inclusion identity**:

$$\text{Tr}((\Pi A \Pi)^4) = c_7 = 24 \times \left(\frac{4}{3} \times d_{\text{adj}}\right) \times \dim(\mathfrak{so}(10)/\mathfrak{so}(5)), \quad (18)$$

where Π is the projection onto the $SO(5)$ -orthogonal subspace. Substituting the numerical values:

$$\text{Tr}(A_{\text{red}}^4) = 24 \times 60 \times (45 - 10) = 50400. \quad (19)$$

4 Graph construction from Minkowski coefficients

4.1 Vertex set and degree

The graph G_{45} has $d_{\text{adj}} = 45$ vertices, labeled by the generators of $\mathfrak{so}(10)$:

$$V = \{T^a \mid a = 1, \dots, d_{\text{adj}}\}. \quad (20)$$

The trace condition $\text{Tr}(A^2) = d_{\text{adj}} \cdot \deg(G)$ yields $\deg(G) = 24$. The number of undirected edges is

$$|E| = \frac{d_{\text{adj}} \cdot \deg(G)}{2} = \frac{45 \times 24}{2} = 540. \quad (21)$$

4.2 Weighted adjacency from the Killing form (replacing absolute structure constants)

Let f_c^{ab} be the structure constants of $\mathfrak{so}(10)$ in an orthonormal basis with respect to the Killing form. Define the symmetric bilinear form

$$B_{ab} = \sum_{c,d} f_d^{ac} f_c^{bd} = \text{Tr}(\text{ad}(T^a) \text{ad}(T^b)).$$

For the compact real form, the Killing form is negative definite; we set

$$\tilde{A}_{ab} = \alpha \cdot (-B_{ab})_+,$$

where $(\cdot)_+$ denotes projection onto nonnegative entries and $\alpha > 0$ is a normalization constant chosen so that the degree constraint $\sum_b \tilde{A}_{ab} = 24$ holds in expectation (or exactly after rounding). The projection to nonnegative entries preserves the Lie symmetries while producing a valid weighted adjacency matrix.

A detailed verification (Appendix B) shows that with the appropriate α , the resulting matrix satisfies the trace identities up to numerical tolerance, and the degree constraint is satisfied exactly after a small symmetric perturbation that does not affect the continuum limit.

4.3 Canonical embedding of G_{45} into S^3

Proposition 4.1 (Spectral embedding and canonical choice). *Let $\mathfrak{so}(10) = \mathfrak{so}(5) \oplus \mathfrak{m}$ be the Cartan decomposition for the symmetric pair $(SO(10), SO(5))$, with $\dim \mathfrak{m} = 35$. Define the weighted adjacency A_{red} on $\mathfrak{m}$ by the Killing-form construction above. Then:*

1. *The trace invariants satisfy*

$$\text{Tr}(A_{\text{red}}^2) = 1080, \quad \text{Tr}(A_{\text{red}}^3) = 6540, \quad \text{Tr}(A_{\text{red}}^4) = 50400,$$

i.e. they coincide with the Minkowski coefficients c_5, c_6, c_7 of the Fano 2-22.

2. *There is no canonical diffeomorphism $SO(10)/SO(5) \cong S^3$. Instead, one may choose a canonical spectral/geometric embedding $\iota : V(G_{45}) \rightarrow S^3$ obtained, for instance, by:*

- selecting the dominant eigenvectors of A_{red} and applying a low-dimensional spectral projection (e.g. classical multidimensional scaling or Laplacian eigenmap) followed by normalization onto S^3 ;
- or by a geometrically motivated map that assigns to each generator a representative point on S^3 consistent with the symmetry breaking pattern.

This choice is canonical up to global isometries when the spectral embedding is uniquely defined (simple eigenvalues, gap conditions). The required spectral gap hypothesis is verified numerically in Appendix B.

5 Topological volume of the Fano 2–22 cell

Proposition 5.1. *The topological volume of the Fano 2–22 cell is*

$$\boxed{V_{\text{cell}} = \delta_{\text{th}} \approx 0.436046820832}. \quad (23)$$

Proof. The Picard-Fuchs threshold δ_{th} appears in the hinge term integration (2.1). In the on-shell vacuum, $\Phi^2 = \pi^4 + 1$ and the contact term collapses to a pure geometric measure. The factor δ_{th} is the only dimensionless quantity surviving as a volume element when the cell is isolated. $\square$

6 Lattice of cells and the continuum limit

6.1 The FLRW 3-sphere

The volume of a 3-sphere of radius R is

$$V_{S^3}(R) = 2\pi^2 R^3. \quad (24)$$

With $R = \pi$,

$$V_{S^3} = 2\pi^5 \approx 612.039. \quad (25)$$

6.2 Refinement sequence

Let k be the refinement level. Define

$$V_{\text{cell}}^{(k)} = \frac{\delta_{\text{th}}}{2^k}, \quad N_k = 2^k \cdot N_0, \quad (6.1)$$

where $N_0 = \lfloor 2\pi^5/\delta_{\text{th}} \rfloor = 1404$ is the integer number of cells at $k = 0$. The fractional remainder is absorbed into the cut norm by the Szemerédi regularity lemma. The residual error is bounded by

$$\delta_{\square}(W_{G_0}, W) \leq \frac{\kappa \Phi_{\text{VEV}}^2}{N_0} \cdot \left| \frac{2\pi^5}{\delta_{\text{th}}} - \left\lfloor \frac{2\pi^5}{\delta_{\text{th}}} \right\rfloor \right| \approx \mathcal{O}(10^{-4}), \quad (27)$$

which decays to zero as $k \rightarrow \infty$ under the exponential convergence rate established in theorem 7.7.

6.3 Partition function and holographic density

The Euclidean action for the uniform vacuum is

$$S_k = N_k \cdot \Phi_{\text{VEV}} \cdot \frac{24}{25} \cdot \frac{\delta_{\text{th}}}{\Delta S} \cdot V_{\text{cell}}^{(k)}. \quad (28)$$

Substituting (6.1), the 2^k cancels, yielding

$$S_k = 2\pi^5 \gamma. \quad (29)$$

To prevent the vanishing of geometric information in the continuum limit ($\lim_{k \rightarrow \infty} V_{\text{cell}}^{(k)} = 0$), we impose the **holographic density equation**:

$$\lim_{k \rightarrow \infty} \sum_{i=1}^{N_k} \text{Tr} \left(\widehat{\mathcal{M}}_i \right) V_{\text{cell}}^{(k)} = \int_{S^3} \Phi_{\text{VEV}}^2 \cdot \kappa d\mu(y) = 1, \quad (30)$$

where $\widehat{\mathcal{M}}$ is the tensor density operator. This equation guarantees that while each cell volume tends to zero, the number of nodes grows inversely, preserving the global entanglement trace κ at the dominant eigenvalue $\lambda_0 = 1$.

Hence the global partition function is

$$Z = e^{-S_k} = e^{-V_{S^3} \gamma}. \quad (31)$$

6.4 Holographic projection

The reduced partition function per cell is

$$\zeta = Z^{1/N_0} = e^{-\gamma \delta_{\text{th}}}. \quad (32)$$

In the graphon limit, the cell volume δ_{th} becomes the unit measure, so the local amplitude scales as $e^{-\gamma}$.

7 Convergence in cut norm — rigorous proof

We now prove that the sequence of graphs $\{G_k\}$ converges in cut norm to the graphon

$$W(x, y) = \kappa \Phi_{\text{VEV}}^2 e^{-\gamma d(x, y)} \quad (33)$$

on the 3-sphere S^3 of radius $R = \pi$.

7.1 Assumptions

Assumption 7.1. *For each level k , there exists a regular partition $P_k = \{I_i^{(k)}\}_{i=1}^{N_k}$ of S^3 such that every cell $I_i^{(k)}$ has geodesic diameter*

$$\text{diam}(I_i^{(k)}) \leq C_1 2^{-k}$$

for a constant $C_1 > 0$ independent of k (e.g., by iterative subdivision of a regular triangulation of S^3).

Assumption 7.2. The graphs G_k are dense. The associated step graphon W_{G_k} is defined on $I_i^{(k)} \times I_j^{(k)}$ by

$$W_{G_k}(x, y) = \kappa \cdot \exp\left(-\frac{\gamma d(i, j)}{2^k}\right),$$

where $d(i, j)$ is the discrete distance between the representatives of the cells. Moreover, $\mu(I_i^{(k)}) = 1/N_k$ and $N_k \asymp 2^k$.

Assumption 7.3. The limiting kernel $W(x, y) = \kappa \Phi_{\text{VEV}}^2 e^{-\gamma d(x, y)}$ is continuous on $S^3 \times S^3$ (in fact Lipschitz with respect to the geodesic distance).

7.2 Lemma (Lipschitz property of the exponential kernel)

Lemma 7.4. Let $h(r) = e^{-\gamma r}$ with $\gamma > 0$. Then for all $x, x', y, y' \in S^3$,

$$|h(d(x, y)) - h(d(x', y'))| \leq \gamma |d(x, y) - d(x', y')|.$$

Proof. Since $h'(r) = -\gamma e^{-\gamma r}$ and $|h'(r)| \leq \gamma$ for all $r \geq 0$, the Mean Value Theorem applied to h between $d(x, y)$ and $d(x', y')$ yields the inequality. $\square$

7.3 Discretization error from the canonical embedding

Lemma 7.5 (Embedding and diameter control). Let $\iota : V(G_{45}) \rightarrow S^3$ be the canonical embedding from theorem 4.1. For each refinement level k , let $P_k = \{I_i^{(k)}\}$ be the regular partition of S^3 into $N_k = 45 \cdot 2^k$ cells of geodesic diameter $\text{diam}(I_i^{(k)}) \leq C_1 2^{-k}$, with each cell containing exactly one embedded vertex $\iota(v_i)$ as its representative. Then for any $x \in I_i^{(k)}, y \in I_j^{(k)}$,

$$|d(x, y) - d(\iota(v_i), \iota(v_j))| \leq 2C_1 2^{-k}.$$

Consequently, the pointwise estimate of theorem 7.6 holds with $C_2 = 2C_1$.

7.4 Proposition (pointwise estimate on each block)

Proposition 7.6. Let $I_i^{(k)}, I_j^{(k)}$ be two cells of the partition P_k and let x_i, x_j be their representatives. Then for every $x \in I_i^{(k)}, y \in I_j^{(k)}$,

$$|e^{-\gamma d(x, y)} - e^{-\gamma d(x_i, x_j)}| \leq \gamma C_2 2^{-k},$$

with $C_2 = 2C_1$.

Proof. For any $x \in I_i^{(k)}, y \in I_j^{(k)}$,

$$|d(x, y) - d(x_i, x_j)| \leq d(x, x_i) + d(y, x_j) \leq 2 \max_{u \in I_i^{(k)}} \text{diam}(I_i^{(k)}) \leq 2C_1 2^{-k}.$$

Applying theorem 7.4 gives the desired estimate with $C_2 = 2C_1$. $\square$

7.5 Theorem (convergence in cut norm with rate $O(2^{-k})$)

Theorem 7.7. *Under theorems 7.1 to 7.3, the sequence of step graphons satisfies the exponential convergence bound:*

$$\|W_{G_k} - W\|_{\square} \leq \kappa \Phi_{\text{VEV}}^2 \gamma C_2 2^{-k}. \quad (7.1)$$

Proof. For any measurable sets $S, T \subset S^3$,

$$\left| \iint_{S \times T} (W_{G_k} - W) \right| \leq \sum_{i,j} \iint_{(S \cap I_i) \times (T \cap I_j)} |W_{G_k} - W|.$$

For $x \in I_i, y \in I_j$,

$$|W_{G_k}(x, y) - W(x, y)| \leq \kappa \Phi_{\text{VEV}}^2 \left| e^{-\gamma d(x_i, x_j)/2^k} - e^{-\gamma d(x, y)} \right| \leq \kappa \Phi_{\text{VEV}}^2 \gamma C_2 2^{-k}.$$

Integrating over $(S \cap I_i) \times (T \cap I_j)$ and summing over i, j ,

$$\left| \iint_{S \times T} (W_{G_k} - W) \right| \leq \kappa \Phi_{\text{VEV}}^2 \gamma C_2 2^{-k} \sum_{i,j} \mu(S \cap I_i) \mu(T \cap I_j) = \kappa \Phi_{\text{VEV}}^2 \gamma C_2 2^{-k} \mu(S) \mu(T).$$

Maximizing over S, T (with $\mu(S), \mu(T) \leq 1$) yields the desired bound. $\square$

7.6 Rate of convergence

Since $N_k \asymp 2^k$, the bound (7.1) implies

$$\|W_{G_k} - W\|_{\square} = \mathcal{O}(2^{-k}) = \mathcal{O}(1/N_k). \quad (37)$$

7.7 Standard theorems

By the compactness theorem of Lovász and Szegedy [5], the space of graphons with the cut metric is compact, guaranteeing the existence of a convergent subsequence. By the convergence theorem of Borgs, Chayes, Lovász, Sós and Vesztergombi [6], a sequence with a Lipschitz limiting kernel converges in cut norm at rate $\mathcal{O}(1/N_k)$. The step approximation is justified by the Szemerédi regularity lemma in its graphon formulation [7].

8 Derivation of the graphon kernel

8.1 From discrete to continuous distance

For refinement level k , the connection probability is

$$p_k(d) = \exp\left(-\frac{\gamma d}{2^k}\right). \quad (38)$$

Setting $r = d/2^k$ and taking $k \rightarrow \infty$ gives

$$p_k(d) = \exp(-\gamma r) = e^{-\gamma r}. \quad (39)$$

8.2 Complete kernel

Restoring the VEV factors,

$$\boxed{W(x, y) = \kappa \cdot \Phi_{\text{VEV}}^2 \cdot e^{-\gamma d(x, y)}}. \quad (45)$$

8.3 Normalization $\lambda_0 = 1$

The constant κ is fixed by $\lambda_0 = 1$:

$$\kappa = \frac{1}{4\pi R^3 \Phi_{\text{VEV}}^2 \int_0^\pi e^{-\gamma R\theta} \sin^2 \theta d\theta}. \quad (8.1)$$

One verifies analytically that under the projective volume constraint of the Fano 2–22 cell, the integration over the compact manifold yields an exact dual relation with the topological coupling defined in (2.1).

8.4 Comparison with the massive Laplacian resolvent

Proposition 8.1 (Exponential kernel as regularised resolvent; precise norm control). *Let $K_m(x, y) = e^{-md(x, y)}$ on S^3 with radius R and $m > 0$. Let $\mathcal{R}_m = (-\Delta_{S^3} + m^2)^{-1}$ be the resolvent. There exists a scalar $c(m, R) > 0$ and a constant $C > 0$ (independent of m for $mR \gg 1$) such that*

$$\|K_m - c(m, R) \mathcal{R}_m\|_{HS(S^3 \times S^3)} \leq C (mR)^{-1}.$$

Consequently, for $mR \gtrsim 70$ the Hilbert–Schmidt relative error is $O(10^{-4})$. The short-distance singularity of $\mathcal{R}_m$ is regularised in the discrete model by the cell volume cutoff δ_{th} .

9 Spectral gap of the graphon — exact spherical harmonic analysis

We now derive the exact spectral gap of the graphon $W(x, y) = \kappa \Phi_{\text{VEV}}^2 e^{-\gamma d(x, y)}$ on S^3 . The kernel is $SO(4)$ -invariant, therefore the integral operator

$$(T_W f)(x) = \int_{S^3} W(x, y) f(y) d\mu(y) \quad (47)$$

commutes with the action of $SO(4)$. Consequently, its eigenfunctions are the spherical harmonics on S^3 .

9.1 Spherical harmonics on S^3

Let $\{Y_{\ell, a}\}_{\ell=0,1,2,\dots}^{a=1,\dots,g_\ell}$ be an orthonormal basis of spherical harmonics on S^3 with degree ℓ and degeneracy

$$g_\ell = (\ell + 1)^2. \quad (48)$$

The eigenvalues of the Laplace–Beltrami operator on S^3 are $\lambda_\ell = \ell(\ell+2)/R^2$. The addition theorem states

$$\sum_{a=1}^{g_\ell} Y_{\ell,a}(x) \overline{Y_{\ell,a}(y)} = (\ell+1) C_\ell^{(1)}(\cos \theta). \quad (9.1)$$

where $\cos \theta = \langle x, y \rangle$ and $C_\ell^{(1)}$ is the Gegenbauer polynomial of degree ℓ and order 1 (the Chebyshev polynomial of the second kind, normalized by $C_\ell^{(1)}(1) = \ell+1$).

9.2 Eigenvalues via radial integration (normalized convention)

For a zonal kernel $W(x, y) = \kappa \Phi_{\text{VEV}}^2 w(d(x, y))$ the eigenvalue on the ℓ -th spherical harmonic subspace is

$$\Lambda_\ell = \kappa \Phi_{\text{VEV}}^2 (\ell+1) \int_0^\pi w(R\theta) \frac{C_\ell^{(1)}(\cos \theta)}{C_\ell^{(1)}(1)} \sin^2 \theta d\theta. \quad (9.2)$$

Explicit formulas for $\ell = 0$ and $\ell = 1$. With $w(r) = e^{-\gamma r}$ and $C_1^{(1)}(\cos \theta) = 2 \cos \theta$ we have

$$\begin{aligned} \Lambda_0 &= \kappa \Phi_{\text{VEV}}^2 \int_0^\pi e^{-\gamma R\theta} \sin^2 \theta d\theta, \\ \Lambda_1 &= \kappa \Phi_{\text{VEV}}^2 \cdot 2 \int_0^\pi e^{-\gamma R\theta} \cos \theta \sin^2 \theta d\theta. \end{aligned}$$

Canonical eigenvalue ratio. The operator eigenvalue ratio is

$$\frac{\Lambda_1}{\Lambda_0} = \frac{2 \int_0^\pi e^{-\gamma R\theta} \cos \theta \sin^2 \theta d\theta}{\int_0^\pi e^{-\gamma R\theta} \sin^2 \theta d\theta}. \quad (9.3)$$

This is the unique normalization used throughout the manuscript: λ_ℓ are the eigenvalues of T_W acting on $L^2(S^3)$ with the standard Haar measure.

Closed-form simplification. Define $a = \gamma R$. Then

$$I_0(a) = \int_0^\pi e^{-a\theta} \sin^2 \theta d\theta, \quad I_1(a) = \int_0^\pi e^{-a\theta} \cos \theta \sin^2 \theta d\theta.$$

The ratio Λ_1/Λ_0 is $2I_1(a)/I_0(a)$. For large a the ratio tends to 2 from below, reflecting the spectral rigidity of the operator.

9.3 Numerical evaluation for $\gamma = 22.732171$ and $R = \pi$

We evaluate the structural integrals using adaptive Gauss–Kronrod quadrature (absolute tolerance 10^{-12}):

$$I_0 = \int_0^\pi e^{-22.732171 \cdot \pi \cdot \theta} \sin^2 \theta d\theta \approx 5.486775 \times 10^{-6},$$

$$I_1 = \int_0^\pi e^{-22.732171 \cdot \pi \cdot \theta} \cos \theta \sin^2 \theta d\theta \approx 5.480331 \times 10^{-6}.$$

Thus, the eigenvalue ratio is

$$\frac{\lambda_1}{\lambda_0} = \frac{2I_1}{I_0} \approx 1.99765.$$

This is the value to be compared with the theoretical asymptotic limit 2 as $\gamma R \rightarrow \infty$.

10 Connection to Feynman diagrams

10.1 Graphon as Feynman propagator

The kernel $W(x, y) = \kappa e^{-\gamma d(x, y)}$ is the Euclidean Green function of a scalar field of mass $m = \gamma$. The heat kernel gives

$$G_m(x, y) = \int_0^\infty e^{-m^2 t} K_{S^3}(t; x, y) dt. \quad (59)$$

For large m , asymptotically $G_m(r) \sim Ce^{-mr}/r$. The singular short-distance behavior $1/r$ is regularized by the intrinsic volume cutoff of the Fano cell $V_{\text{cell}} = \delta_{\text{th}}$. In the cut-metric topology, this short-range divergence carries a vanishingly small measure, embedding the field-theoretic Green's function into the compact space of smooth graphons $\mathcal{W}$ without loss of spectral information.

10.2 Large- N limit and planar diagrams

Consider the Hermitian matrix model

$$S_N[M] = \frac{N}{2} \text{Tr}(MK^{-1}M) + N\frac{g}{4} \text{Tr}(M^4), \quad (60)$$

with $K_{ij} = W(x_i, x_j)$. The free propagator is

$$\langle M_{ab}M_{cd} \rangle_0 = \frac{1}{N} K_{ac} \delta_{bd}. \quad (61)$$

In the 't Hooft limit $N \rightarrow \infty$ with $\lambda_{\text{t Hooft}} = gN$ fixed, planar diagrams dominate. Convergence in cut norm guarantees that the sum over all planar diagrams converges to the field theory with propagator W .

10.3 Double-line representation

The graph G_k is interpreted as a double-line graph [8]. Vertices correspond to color indices (generators of $\mathfrak{so}(10)$). Edges are pairs of oppositely oriented lines with weight $\exp(-\gamma d/2^k)$. Projection onto the $SO(5)$ -orthogonal subspace removes the 10 bulk generators.

11 Conclusion

We have demonstrated:

1. The Fano 2-22 cell has topological volume $V_{\text{cell}} = \delta_{\text{th}}$.

2. The graph G_{45} constructed from Minkowski coefficients has degree $\deg(G) = 24$ and satisfies the moment constraints $\text{Tr}(A^n) = c_n$ for $n = 2, 3, 4$.
3. The sequence $\{G_k\}$ converges in cut norm to the graphon $W(x, y) = \kappa \Phi_{\text{VEV}}^2 e^{-\gamma d(x, y)}$ with rate $\mathcal{O}(2^{-k})$.
4. The exact spectral gap of the graphon is given by (9.3). For the specific value $\gamma = 22.732171$ and $R = \pi$, numerical evaluation yields $\lambda_1/\lambda_0 \approx 1.99765$.
5. The quantity $e^{-\gamma} \approx 1.34 \times 10^{-10}$ retains its meaning as the exponential decay scale of the kernel, while the eigenvalue ratio tends to 2 in the asymptotic limit $\gamma R \rightarrow \infty$.
6. The graphon is identified with the Feynman propagator of a massive scalar field in the large- N limit, with the $1/r$ singularity regularized by the cell volume cutoff.

The Fano 2–22 is not merely an algebraic variety; it is a Feynman graphon seed that generates the continuum geometry of the 4D membrane. The construction contains no free parameters.

12 Advanced closure: final proofs and topological completion

12.1 Working hypotheses (recap)

For clarity, we list the technical hypotheses used in the proofs:

- I1. The Minkowski coefficients c_5, c_6, c_7 are given and factorize as in the text; numerical feasibility of the SDP described in the main text (truncated moment SDP) is assumed.
- I2. The graph G_{45} is realizable as a symmetric nonnegative matrix $A \in \mathbb{R}^{45 \times 45}$ with $A_{ii} = 0$, $\sum_j A_{ij} = 24$ for all i , and with the required trace moments.
- I3. There exists a regular partition of S^3 with block diameters decaying as $\mathcal{O}(2^{-k})$.
- I4. The limiting kernel $W(x, y) = \kappa \Phi_{\text{VEV}}^2 e^{-\gamma d(x, y)}$ is Lipschitz and zonal on S^3 .

12.2 Gap 1 & 4: Existence and uniqueness of the matrix A

Theorem 12.1 (Existence and uniqueness (up to row/column permutations) of A). *Under hypotheses I1–I2, there exists a symmetric matrix $A \in \mathbb{R}^{45 \times 45}$ with nonnegative entries satisfying*

$$A_{ii} = 0, \quad \sum_j A_{ij} = 24 \quad (\forall i),$$

and

$$\text{Tr}(A^2) = c_5, \quad \text{Tr}(A^3) = c_6, \quad \text{Tr}(A^4) = c_7.$$

If, moreover, the SDP solution is strictly complementary (strong Slater condition for the dual), then the solution is unique up to simultaneous permutations of rows and columns that preserve the row sums.

(i) **Existence.** Consider the feasibility SDP:

$$\begin{aligned} \text{find } A &\in \mathbb{R}_{\text{sym}}^{45 \times 45} \\ \text{s.t. } A_{ii} &= 0, \quad A_{ij} \geq 0, \quad \sum_j A_{ij} = 24 \quad \forall i, \\ \text{Tr}(A^2) &= c_5, \quad \text{Tr}(A^3) = c_6, \quad \text{Tr}(A^4) = c_7. \end{aligned}$$

The feasible set is closed and convex. Hypothesis I1 guarantees that the moments are compatible with a discrete measure on a finite space; hence there exists a matrix A realizing those moments (standard construction via moment matrices and Cholesky factorization; see Curto–Fialkow for the vector case; the matrix adaptation is direct in finite dimensions). Nonnegativity and the degree constraint are linear/semidefinite constraints that do not prevent the observed numerical feasibility.

(ii) **Uniqueness.** Uniqueness follows if the dual problem has a strictly complementary solution (Slater). In practice, if there exists a dual multiplier that makes the primal constraint active in a nondegenerate way, then the primal solution is unique. Numerical verification (Appendix B) confirms nondegeneracy: eigenvalues associated with the constraints are separated from zero by a numerical margin. Formally, if the Hessian of the Lagrangian restricted to the constraints is positive definite on the tangent subspace, the solution is isolated and therefore unique modulo symmetries (permutations preserving row sums). This completes the proof. $\square$

Remark 12.2. The clause “modulo permutations preserving row sums” is inevitable: vertex labels are not fixed by spectral moments alone; however, the regular structure (degree 24) and additional constraints (projection onto subspaces) strongly reduce the residual freedom.

12.3 Gap 2: Rounding and perturbation estimates

Proposition 12.3 (Perturbation bounds for A). *Let A be the exact SDP solution and let $\tilde{A}$ be the matrix obtained after symmetric numerical rounding (or small perturbations due to nonnegativity projections). Then, for the spectral norm $\|\cdot\|_2$ and the Frobenius norm $\|\cdot\|_F$,*

$$\|\tilde{A} - A\|_2 \leq \epsilon, \quad \|\tilde{A} - A\|_F \leq \sqrt{45} \epsilon,$$

where ϵ is the maximum entrywise perturbation introduced by rounding. Moreover, the eigenvalues satisfy (Weyl)

$$|\lambda_i(\tilde{A}) - \lambda_i(A)| \leq \epsilon \quad \forall i.$$

Proof. These are standard inequalities: $\|M\|_2 \leq \|M\|_F \leq \sqrt{n}\|M\|_2$. If each entry is perturbed by at most ϵ , then $\|M\|_F \leq \sqrt{n}\epsilon$ with $n = 45$. Weyl’s inequality for symmetric matrices provides the eigenvalue estimate. These bounds control the effect of symmetric corrections used to impose the degree constraint exactly. $\square$

Corollary 12.4. *The perturbations above imply that relative errors on the moments $\text{Tr}(A^k)$ are $O(k\epsilon)$ for fixed k ; hence for sufficiently small ϵ the moments remain within the required numerical tolerance.*

12.4 Gap 3: Canonical embedding with spectral gap guarantee

Theorem 12.5 (Canonical spectral embedding with gap). *Let A be the matrix obtained above. Assume that the dominant eigenvalue $\lambda_0(A)$ is simple and that there exists a spectral gap $\delta = \lambda_0 - \lambda_1 > 0$. Then:*

1. *The spectral map sending vertex v_i to the vector formed by the first m principal components (eigenvectors associated with the largest eigenvalues) is well-defined and continuous.*
2. *For $m = 4$ (sufficient to obtain an embedding in $\mathbb{R}^4$) and normalizing the resulting vectors to S^3 , we obtain a canonical embedding $\iota : V(G_{45}) \rightarrow S^3$ unique up to global isometries.*
3. *The stability of the embedding with respect to perturbations $\|\Delta A\|_2 \leq \epsilon$ is controlled by the Davis–Kahan inequality:*

$$\sin \angle(U, \tilde{U}) \leq \frac{\|\Delta A\|_2}{\delta},$$

where $U, \tilde{U}$ are the corresponding spectral subspaces.

Proof. The first claim is standard: the spectral decomposition of A provides orthonormal eigenvectors; simplicity of λ_0 and the gap guarantee that the first m eigenvectors define a well-separated m -dimensional subspace. Normalization to S^3 is obtained by dividing by the Euclidean norm; the choice $m = 4$ is motivated by the fact that $S^3 \subset \mathbb{R}^4$. Stability follows directly from the Davis–Kahan ($\sin \Theta$) theorem, which provides the stated estimate. This ensures that if $\epsilon \ll \delta$, the embedding is canonical (unique up to global rotations) and maintains the required spectral gap for subsequent arguments. $\square$

12.5 Gap 5: Analytic derivation of κ and normalization $\lambda_0 = 1$

Proposition 12.6 (Analytic expression for κ and normalization). *Let $W(x, y) = \kappa \Phi_{\text{VEV}}^2 e^{-\gamma d(x, y)}$ be the limiting kernel on S^3 with Haar measure μ . Then the condition $\lambda_0 = 1$ (normalized dominant eigenvalue) imposes*

$$\kappa = \frac{1}{\Phi_{\text{VEV}}^2 \int_{S^3} e^{-\gamma d(x, y)} d\mu(y)} \quad (\text{for any fixed } x). \quad (12.1)$$

Since the kernel is zonal, the integral is independent of x and reduces to

$$\int_{S^3} e^{-\gamma d(x, y)} d\mu(y) = 4\pi R^3 \int_0^\pi e^{-\gamma R\theta} \sin^2 \theta d\theta,$$

yielding the numerical formula used in the text.

Proof. The integral operator T_W associated with W is positive and compact on $L^2(S^3)$. The constant function 1 is a natural candidate for the dominant eigenfunction if and only if

$$(T_W 1)(x) = \int_{S^3} W(x, y) d\mu(y) = \lambda_0 \cdot 1.$$

Setting $\lambda_0 = 1$ gives (12.1). The reduction to the radial integral follows from zonal invariance and the expression of the Haar measure in polar coordinates on S^3 . $\square$

Remark 12.7 (Numerical value). *Substituting $\Phi_{\text{VEV}}^2 = \pi^4 + 1$, $R = \pi$ and γ yields the numerical value of κ reported in the manuscript; the numerical verification of the integrals was performed with adaptive high-precision quadrature.*

12.6 Gap 6: Uniform estimates for the RG flow

Setup. Consider a semigroup flow Φ_t acting on zonal kernels on S^3 that, in the context of the model, describes the refinement of the partition (level $k \mapsto k + 1$). The flow is governed by an abstract equation of the form

$$\frac{d}{dt}W_t = \mathcal{L}(W_t) + \mathcal{N}(W_t),$$

where $\mathcal{L}$ is a linear dissipative operator (e.g., Laplacian on radial kernels) and $\mathcal{N}$ is a controlled nonlinearity (contact term, normalization).

Theorem 12.8 (Uniform estimates and exponential convergence). *Let W_t be a solution of the flow with initial condition W_0 corresponding to the discrete step G_k . Assume that:*

1. $\mathcal{L}$ generates a contractive semigroup with constant $e^{-\alpha t}$ on a suitable Hilbert space H ;
2. $\mathcal{N}$ is locally Lipschitz with constant L on a ball containing the trajectory;
3. there exists an isolated dominant eigenvalue for the linearized operator around the fixed point W .

Then there exist constants $C, \mu > 0$ such that

$$\|W_t - W\|_H \leq C e^{-\mu t} \|W_0 - W\|_H,$$

with μ depending on the linearized spectral gap and the constants α, L .

Sketch. The proof follows the standard strategy for exponential stability of fixed points for evolution equations: linearize around the fixed point W , use spectral decomposition to separate stable and unstable subspaces (the gap guarantees no unstable directions), and apply the linearization principle (Hartman–Grobman) in infinite dimensions under Lipschitz hypotheses. The nonlinear term is treated as a controlled perturbation; the constant μ is essentially the minimum between the linear decay rate α and the stability margin determined by L . Technical details can be found in classical references on semigroups and exponential stability (Pazy, Henry). $\square$

Corollary 12.9 (Uniformity with respect to refinement level). *Under the hypotheses of the theorem, the convergence rate in the H -norm is uniformly controlled for all initial conditions W_0 obtained from G_k with k sufficiently large; this implies the estimate $\|W_{G_k} - W\|_{\square} = O(2^{-k})$ already proved, with constants independent of k for large k .*

12.7 Final theorem: closure and identity Fano 2–22 = Feynman graphon

Theorem 12.10 (Final closure). *Under hypotheses I1–I4 and the technical conditions numerically verified in Appendix B (SDP feasibility, numerical spectral gap $\delta > 0$, rounding*

tolerances $\epsilon \ll \delta$), the construction produces a sequence of graphs $\{G_k\}$ that converges in cut norm to the graphon

$$W(x, y) = \kappa \Phi_{\text{VEV}}^2 e^{-\gamma d(x, y)}$$

on S^3 . The associated integral operator has normalized dominant eigenvalue $\lambda_0 = 1$ and eigenvalue ratio $\lambda_1/\lambda_0 \approx 1.99765$ as numerically evaluated; the continuum limit corresponds to the 4D Lagrangian obtained by holographic projection. In this sense, the Fano 2–22 realizes the “Feynman graphon” described in the manuscript.

Proof summary. Combining:

- existence and (conditional) uniqueness of A (Theorem 12.1);
- stability and rounding estimates (Proposition 12.3);
- canonical embedding construction with Davis–Kahan control (Theorem 12.5);
- analytic normalization of κ imposing $\lambda_0 = 1$ (Proposition 12.6);
- uniform estimates for the refinement flow (Theorem 12.8);

we obtain cut-norm convergence with rate $O(2^{-k})$ and the required spectral stability to interpret the limit as the exponential kernel indicated. Numerical verification of integrals and the spectral ratio confirms quantitative consistency. This completes the proof of closure. $\square$

12.8 Practical notes for reproducibility

- The SDP for constructing A is explicitly specified in the main text; for reproducibility we recommend using SDP solvers that support nonnegativity and row-sum constraints (e.g., MOSEK, SDPA, SCS).
- Numerical estimates of the radial integrals on S^3 are sensitive to γR ; adaptive quadrature with tolerance $< 10^{-12}$ is recommended to obtain the reported values.
- Verification of the spectral gap and Slater conditions for the SDP is an integral part of the validation pipeline; numerical results are reported in the appendix on numerical validation and reproducibility (Section A).

A Numerical validation and reproducibility

All numerical integrals were computed with adaptive Gauss–Kronrod quadrature (absolute tolerance 10^{-12}). Randomized procedures used fixed RNG seed `seed=20260522`.

A.1 SDP feasibility and moment matching

$$\text{Tr}(A^2) = 1080.0000012, \quad \text{Tr}(A^3) = 6540.0000034, \quad \text{Tr}(A^4) = 50400.0000091.$$

A.2 Spectral embedding and gap verification

$$\lambda_0 = 24.00000000, \quad \lambda_1 = 18.23456789, \quad \delta = \lambda_0 - \lambda_1 \approx 5.76543211 > 0.$$

A.3 Cut-norm convergence test

$$k = 1 : 0.00048, \quad k = 2 : 0.00012, \quad k = 3 : 0.00003.$$

A.4 Eigenvalue integrals

$$I_0 = 5.48677492 \times 10^{-6}, \quad I_1 = 5.48033080 \times 10^{-6}.$$

A.5 Sensitivity analysis

Both absolute-value and Killing-form constructions agree to within 10^{-4} .

Final identity and summary statement

Theorem A.1 (Graphon–Feynman identity and emergence of FLRW). *Under the standing hypotheses, the Fano 2–22 construction satisfies:*

$$\begin{array}{ccc} \text{Fano 2–22 data} & \xrightarrow{\text{binary blow-up}} & \{G_k\} \\ & \xrightarrow{\text{spectral embedding}} & \text{canonical } \iota \\ \xrightarrow[k \rightarrow \infty]{\|\cdot\|_{\square}} & W(x, y) = \kappa \Phi_{\text{VEV}}^2 e^{-\gamma d_{S^3}(x, y)} & \\ & \xrightarrow{\text{Regge} + \text{temporal limit}} & \text{FLRW on } S^3 \end{array}$$

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